# EMACS, THE ULTIMATE

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*National University of Singapore*

*Singapore, October 27th, 2017*

Which editor do you use the most?

# CASE STUDY: REAL PROGRAMMERS

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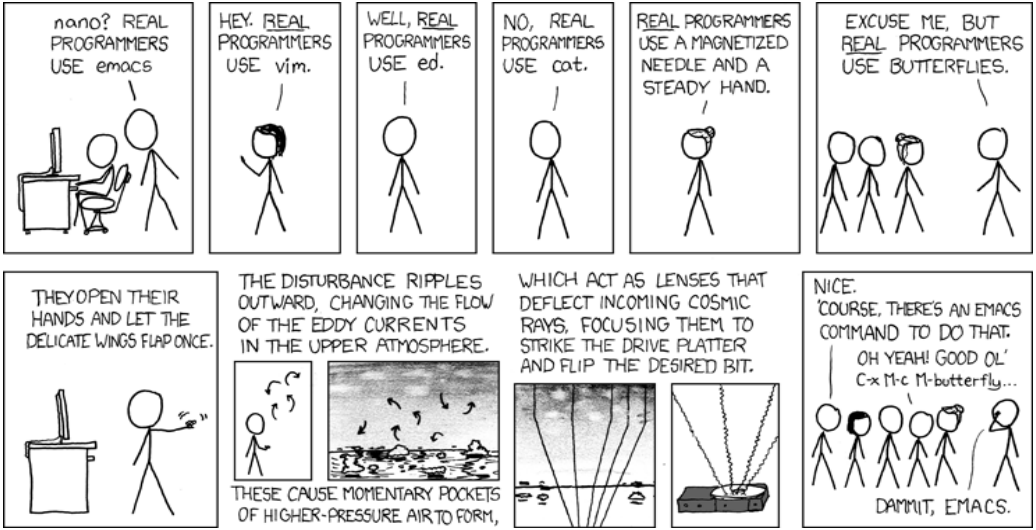


https://xkcd.com/378/

# M-x butterfly

# Outline


1. Emacs, the basic
2. Emacs, the advance
3. Emacs, the ultimate
4. Getting started
5. Conclusion

# A few notes

- The talk title is borrowed from: `lambda-the-ultimate.org`
- Introduce what I know that Emacs can do
- No intention to compare Emacs with other editors/IDEs
- Feel free to interrupt me for questions, demo requests, etc

# Emacs, the basic

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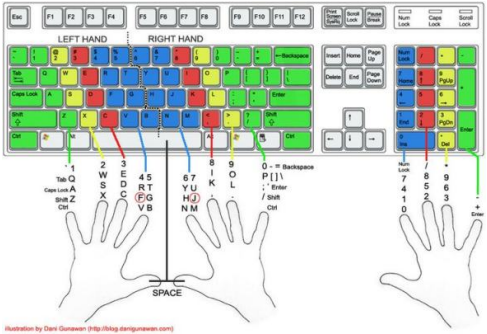

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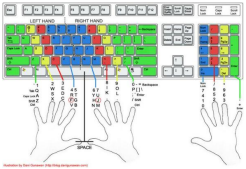

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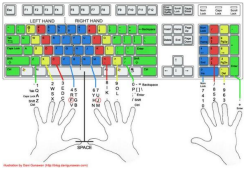


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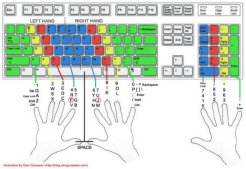


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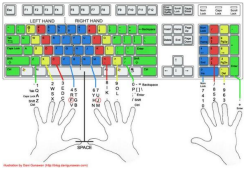


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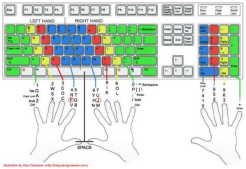


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$\Rightarrow$ Emacs helps to avoid these problems!

# What is Emacs?

- A text editor born in 1976 and still lives actively today
- The name Emacs stands for Editor MACroS
- The first Emacs version was written in 1976 by David A. Moon and Guy L. Steele Jr
- The most popular version GNU Emacs was created by Richard Stallman in 1984
- Many modern editors such as Eclipse, IntelliJ IDEA, Sublime, Visual Studio have an Emacs simulation mode

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- Additional benefit: Unix/Linux Shell uses the same movement keys like Emacs!

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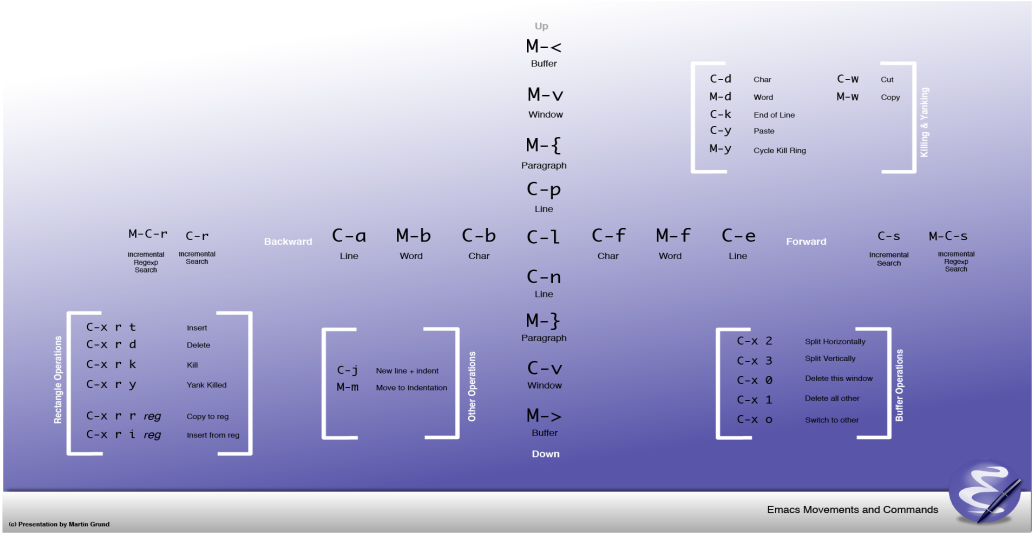


By Martin Grand

# Buffer and window

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- Emacs buffer:
  - The Emacs object containing text
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- Emacs window:
  - The Emacs object showing a buffer
  - Emacs' window $\neq$ window in Linux/Windows (Emacs calls it *frame*)
  - `C-x 0, 1, 2, 3`: delete, maximize, split horizontally/vertically
  - `C-x {, }, ^, _`: shrink, enlarge horizontally/vertically

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- Kill ring
  - The greedy clipboard of Emacs
  - `M-w`, `C-w`, `C-y`: copy, cut, yank (paste)
  - `M-y`: paste the previously copied/cut texts
  - `C-k`, `M-d`, `C-Backspace`, `C-Del`, ... store deleted texts into kill ring

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- Block editing: `C-x SPC`
- Text undo:
  - Default undo: greedy, remember everything: undo of undo of undo ...
  - Modern undo: `undo-tree` with visualization

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  - `M-l`, `M-u`, `M-c`: **l**owercase, **u**ppercase, **c**apitalize first character
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- Text search, replace and highlighting in current buffer:
  - Search: `isearch`, `swivel`, `occur`
  - Replace: `anzu`
  - Highlight: `highlighting`
  - Regular expression supported

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- Buffer navigation: `helm-mini`, `previous/next buffer`
- Working with other-window: `<command>-other-window`
- Others: `just-one-space`, `delete-blank-lines`, `repeat`

# Understand Emacs' components

- Buffer, window, frame
- Major mode: display buffer, visualize and manipulate content
- Minor mode: auxiliary editing and visualizing features
- Mode line, header line: all auxiliary information
- Mini-buffer, echo area: user interacts with Emacs commands

# Explore Emacs' documentation

- Describe major mode: `C-h m` (`M-x describe-mode`)
- Describe key binding: `C-h k` (`M-x describe-key`)
- Describe function: `C-h f` (`M-x describe-function`)
- Describe variable: `C-h v` (`M-x describe-variable`)

# Emacs, the ultimate

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- *Weakness*: code refactoring is limited

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- Text diff and merge: `diff`, `ediff`

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- More handy, useful features, and even game, screensaver!

# Getting started

# Emacs Starter Kits

| Name | Stars | Forks | Contri-butors | Commits | GitHub Sites |
|---|---|---|---|---|---|
| **Spacemacs** | 12.430 | 3.219 | 574 | 7.405 | `GH/syl20bnr/spacemacs` |
| **Purcell emacs** | 3.668 | 1.866 | 16 | 2.638 | `GH/purcell/emacs.d` |
| **Prelude** | 3.470 | 1.453 | 120 | 1.239 | `GH/bbatsov/prelude` |
| **Emacs live** | 1.240 | 245 | 34 | 703 | `GH/overtone/emacs-live` |
| **Magnars emacs** | 1.135 | 206 | 8 | 1.402 | `GH/magnars/.emacs.d` |

Updated in August 05$^{th}$, 2017

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- Useful Emacs packages and libraries:
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- Write your own Emacs functions or packages:
  - Need to be familiar with Emacs Lisp language
  - Starting point: copy an existing function and modify it

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  - Solution: positive thinking – it could be fun!

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